

2019



Group B [6x5=30]

Attempt any SIX questions.

2. Differentiate Object Oriented Programming and Object Based Programming. Explain some of the major features of C# language. [2+3]
3. Explain overview of Microsoft .NET framework and its components in detail. [5]
4. What do you mean by property in C# language? How is it different from method? Compare automatic property with other types of property with suitable example. [1+1+3]
5. Define constructor. Explain different types of constructors used in C# with example. [1+4]
6. Define inheritance. Write a C# program to demonstrate multilevel and multiple inheritance. [1+2+2]
7. What is generics? List different types of generic classes. Explain delegate with example. [1+1+3]
8. What do you mean by lambda expression? Explain different types of lambda expression used in C# with example. [1+4]

Group C

Attempt any TWO questions. [2x10=20]

9. a) Write a program to create user registration form in one ASP.NET web page and display filled data in another page. [7]
b) Write a program for handling exception in ASP.NET. [3]
10. a) How virtual method is used to achieve dynamic binding in C#? Explain with the help of suitable program. [1+4]
b) Define operator overloading. Write a C# program to overload binary operator. [1+4]
11. a) What is LINQ? Write a program to select employees whose salary is greater than 20000 and whose address is Kathmandu using LINQ. [1+4]
b) Write a C# program to show insert and select operation in database. [5]

Course Title: **DotNet Technology (3 Cr.)**

Course Code: **CACS302**

Year/Semester: **III/V**

Class Load: **6 Hrs. / Week (Theory: 3 Hrs., Practical: 3 Hrs.)**

Course Description:

This course covers different concepts of .NET framework. It also covers basic to advanced features of C# language including language basics, creating types and inheritance, delegates, events, lambda expressions, LINQ, working with databases, and developing web applications using ASP.NET.

Course Objectives:

The primary objective of this course is to provide concepts of .NET framework and different concepts of C# programming language and make students familiar with their uses and applications.

Course Contents:

Unit 1: Introducing C# and the .NET Framework (7 Hrs.)

Object Orientation; Type Safety; Memory Management; Platform Support; C# and CLR; CLR and .NET Framework; Other Frameworks; Framework Overview; .NET Standard 2.0; Applied Technologies

Unit 2: The C# Language Basics (12 Hrs.)

Writing Console and GUI Applications; Identifiers and Keywords; Writing Comments; Data Types; Expressions and Operators; Strings and Characters; Arrays; Variables and Parameters; Statements (Declaration, Expression, Selection, Iteration, and Jump Statements); Namespaces

Unit 3: Creating Types in C# (12 Hrs.)

Classes; Constructors and Deconstructors; this Reference; Properties; Indexers; Static Constructors and Classes; Finalizers; Dynamic Binding; Operator Overloading; Inheritance; Abstract Classes and Methods; base Keyword; Overloading; Object Type; Structs; Access Modifiers; Interfaces; Enums; Generics

Unit 4: Advanced C# (14 Hrs.)

Delegates; Events; Lambda Expressions; Exception Handling; Introduction of LINQ; Working with Databases; Writing Web Applications using ASP.NET

Laboratory Work: The laboratory work includes writing console and/or GUI programs in C#

- To implement basic language features
- To create classes and objects and to implement different object-oriented features
- To implement inheritance
- To implement advanced features like delegates, event handling, lambda expressions, exception handling
- To implement LINQ and database applications